

Plan

- Examples of optimization problems
- Examples of Linear Programming Problems (LPP)
- Solution of an LPP by the Graphical Method
- Extreme points and Corner points
- Exercises and Questions

Optimization Problems

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- **Extremal principles of nature:**
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Nature breaks a ray of light at equal angles, if it does not unnecessarily want it to meander to no purpose.
- **Laws of refraction of light.**
What characterizes the trajectory of light moving from one point to another in a non homogeneous medium is that it is traversed in a minimum time.

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Linear Programming Problems

Diet Problem: For U.S soldiers, World War II

- Let there be m nutrients $N_1, N_2, \dots, N_m$ and n food products, $F_1, F_2, \dots, F_n$, available in the market which can supply these nutrients.

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- To decide on a diet of minimum cost consisting of the n food products (in various quantities) so that one gets the required amount of each of the nutrients.

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- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of (P) to be transported from S_i to D_j such that the demands d_j are met and the cost of transportation is minimum.

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Linear Programming Problem

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- The function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$ is called the **objective function**, the constraints $\mathbf{x} \geq \mathbf{0}$ are called the **non negativity constraints** of the LPP.

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- and $\mathbf{c}^T \mathbf{x}$ are all **linear functions** from $\mathbb{R}^n \rightarrow \mathbb{R}$, hence the name **linear programming problem**.

Linear function, Feasible solution, Optimal Solution

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The **feasible region** of a LPP, denoted by $\text{Fea}(\text{LPP})$ is given by,

$$\text{Fea}(\text{LPP}) = \{ \mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}, A\mathbf{x} \leq \mathbf{b} \}.$$

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- If the LPP has an **optimal solution**, then the value of the objective function $\mathbf{c}^T \mathbf{x}_0$ where $\mathbf{x}_0$ is an **optimal solution** of the LPP is called the **optimal value** of the LPP.

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Max $5x + 2y$

subject to

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Optimal solution = $[2, 0]^T$, Optimal value=-2

Hyperplanes, Normals, Closed Half Spaces, Polyhedral set

- A subset H of $\mathbb{R}^n$ is called a **hyperplane** if it can be written as:

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- A collection of hyperplanes $H_1, \dots, H_k$ in $\mathbb{R}^n$ is **Linearly Independent (LI)** if the corresponding normal vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ are linearly independent as vectors in $\mathbb{R}^n$

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 where $\mathbf{O}$ is the zero vector
- If they are not **LD** then the vectors are **LI**, then the only solution to **(**)** is $c_1 = \dots = c_k = 0$

- For $k = 2$, $\mathbf{a}_1, \mathbf{a}_2$ is **LD** $\Leftrightarrow$ there is a solution to **(**)** with either one or both c_1, c_2 nonzero
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- A set consisting of a single non zero vector is **LI**

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Min $-x + 2y$,

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 Normals to H_4 is $\{c[2, 4]^T, c \neq 0\}$
 $\{[1, 2]^T, [2, 4]^T\}$ is **LD** since $2[1, 2]^T - 1[2, 4]^T = \mathbf{0}$.

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- The feasible region of the above problem is **empty**. This problem is called **infeasible**, and since it **does not** have a **feasible solution** it obviously **does not** have an **optimal solution**

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That is, if $\mathbf{x}_1$ and $\mathbf{x}_2$ are two optimal solutions of an LPP, then all $\mathbf{y}$ such that $\mathbf{y} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$, $0 \leq \lambda \leq 1$, are also optimal solutions of the LPP

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 $[\frac{2}{3}, 0]^T = \frac{1}{3}[0, 0]^T + \frac{2}{3}[1, 0]^T$ is a **convex combination** as well as a **strict convex combination** of $\mathbf{x}_1$ and $\mathbf{x}_2$

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The answer to this question is **YES**, due to **Weierstrass**, called the **Extreme Value Theorem**

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- Given an LPP with a nonempty feasible region, $Fea(LPP) = S \subset \mathbb{R}^n$, an $\mathbf{x} \in S$ is called a corner point of S , if $\mathbf{x}$ lies at the point of intersection of n linearly independent defining hyperplanes of S

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- All points on the **boundary** of a **disc** are **extreme points** of the disc
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- **Theorem:** If $S = \text{Fea}(LPP)$ is nonempty, where
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- **Exercise:** Think of a LPP $(m + n)$ constraints such that the number of **extreme points** of the $\text{Fea}(LPP)$ is equal to $(m + n)C_n$

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- **Exercise:** Think of a **convex set** defined by only one hyperplane. Will it have any **extreme point**?